Preserving Rare-Class Signal in Federated Learning

under Statistical and System Heterogeneity

A Report Presented

by

Barbara Koch

Departments

Department of Applied Mathematics and Theoretical Physics
Department of Physics (Cavendish Laboratory)

Degree

MPhil in Data Intensive Science

Supervision

Dr Joshua Kaggie

Peterhouse
University of Cambridge
1st July 2026

Abstract

Federated learning lets institutions train a shared model without pooling private data, but real federations are heterogeneous in both their data and their compute, and medical image datasets are severely class-imbalanced. This thesis studies federated optimisation on DermaMNIST, a seven-class dermatology benchmark dominated by a single benign class, comparing FedAvg, FedProx, and FedNova across controlled statistical- and system-heterogeneity regimes and reporting test macro-F1 at the best-validation checkpoint. Rather than crowning a universal winner, it maps the regimes in which each method preserves or loses rare-class signal. The central finding is that under Li-style straggler asymmetry the FedProx advantage (about +0.115 macro-F1) is carried almost entirely by its γ-inexact acceptance of partial updates rather than by the proximal term, and that FedNova is robust under deterministic learning-rate asymmetry yet collapses under random per-round stragglers. Across the failing regimes, rare lesions are misrouted into the majority class. These results come from controlled DermaMNIST benchmark simulations with $n = 3$ mechanism probes, and map regimes rather than establishing a universal optimiser ranking.

Declaration

This thesis contains 6,862 words.

I declare that this thesis is original and is submitted as part of the requirements for the degree. Except where explicitly stated, it contains nothing which is the outcome of work done in collaboration with others.

I used GitHub for version control and repository management. Claude by Anthropic was used for assistance with coding and debugging, and ChatGPT by OpenAI was used for organisation, planning, and structuring support. The experiments, analysis, interpretation, and final submitted text remain my own responsibility.

Contents

List of Figures

List of Tables

1 Introduction

1.1 Motivation

Diagnostic models improve with large, diverse image collections, but in medicine those collections are fragmented across institutions and cannot usually be pooled centrally because of privacy regulation and governance. Federated learning sidesteps this by keeping each institution's data in place and exchanging only model updates (McMahan et al., 2017; Rieke et al., 2020; Sheller et al., 2020), which makes it a natural setting for collaborative learning across hospitals (Kairouz et al., 2021; Pati et al., 2022). It also introduces two difficulties that are acute in medicine. Under *statistical heterogeneity*, sites see different case mixes, so client data is non-IID (Zhao et al., 2018); under *system heterogeneity*, sites differ in compute and availability, so some clients return only partial updates in a round. The two coincide most damagingly when the under-resourced site is also the one holding the rare cases.

Medical data is, in addition, severely class-imbalanced, a recognised obstacle for federated training (Wang et al., 2021). DermaMNIST (Yang et al., 2023), derived from the HAM10000 dermatoscopic collection (Tschandl et al., 2018) and used throughout this thesis, is dominated by benign melanocytic nevi, while the diagnostically urgent lesions are rare; federated learning has itself been applied to skin-lesion classification (Yaqoob et al., 2023). A model can therefore post high accuracy while failing on exactly the lesions that matter, so we report macro-F1 and per-class rare-class diagnostics rather than accuracy. Crucially, the information needed to recognise a rare lesion reaches the global model only through the updates of the client that holds it; if heterogeneity under-weights or discards those updates, the rare-class signal is lost. The quantity that governs rare-class performance is thus the *flow of rare-client signal into the global model*, and the federated optimiser — not the network architecture — is the lever that protects or forfeits it.

1.2 Research problem

The optimisers proposed for these conditions — FedAvg (McMahan et al., 2017), FedProx with its proximal term and tolerance for inexact updates (Li et al., 2020), and FedNova with its work-normalised aggregation (Wang et al., 2020) — are usually compared by which scores higher on a task (Li et al., 2022), rarely isolating *why*, often conflating the two heterogeneities, and judging by accuracy. This thesis instead asks *when* each method preserves rare-class performance, treating the answer as a *regime map* rather than a universal ranking. It separates four levers that are usually bundled together — the data partition, the optimiser, the partial-update protocol, and loss-side imbalance correction — and varies them one at a time. All experiments are simulated federated learning on a controlled benchmark, so the work maps mechanisms rather than validating a clinical deployment.

1.3 Research questions and contributions

This framing raises four questions. Because the literature often credits FedProx and FedNova with broad gains, the thesis first asks whether statistical heterogeneity alone is enough to separate FedAvg and FedProx or whether additional stress is required; then, where FedProx does help under stragglers, whether that benefit comes from its proximal regularisation or from its γ-inexact acceptance of partial updates; whether FedNova's work-normalisation is robust in general or only under deterministic learning-rate asymmetry, failing under random round-to-round straggling; and finally whether rare-class collapse into the majority class is the shared failure mode, and whether the FedProx advantage is merely class-imbalance correction in disguise. Chapter 3 follows this order: statistical heterogeneity (§3.1), the proximal coefficient (§3.2), straggler asymmetry (§3.3), FedNova regime-dependence (§3.4), and rare-class collapse with the loss-side comparison (§3.5).

Answering these yields the contributions of the thesis. It maps the regimes in which FedAvg, FedProx, and FedNova preserve or lose rare-class signal, and decomposes the FedProx straggler-regime advantage into its proximal and γ-inexact components using matched class-disjoint (L4) and quantity-only (L1) partitions that fix when the effect can appear. It then characterises FedNova's regime-dependence across deterministic and random system heterogeneity, and identifies rare-class collapse into the dominant melanocytic-nevi class as the common failure mode. Finally, through the same L4/L1 decomposition, it shows the FedProx straggler-regime advantage to be a drift-and-protocol effect, with weighted cross-entropy a competitive loss-side alternative rather than evidence that the advantage is merely class-imbalance correction in disguise.

2 Methods

2.1 Problem setup and design

We consider a federation of K clients coordinated by a central server that together train a single global model without exchanging raw data (Figure 2.1). Client i holds a private training set D_i of n_i examples, with $N = \sum_i n_i$. In each round t the server broadcasts the global model w^t; every participating client runs E epochs of local SGD on D_i to obtain w_i^{t+1} and returns its update

$$d_i = w_i^{t+1} - w^t.$$

The baseline rule, FedAvg (McMahan et al., 2017), recombines these by the sample-weighted mean

$$w^{t+1} = \sum_i \frac{n_i}{N} w_i^{t+1}.$$

FedProx (Li et al., 2020) and FedNova (Wang et al., 2020) modify, respectively, the local objective and the way the updates d_i are normalised before combination (§2.4.1).

The aim is not to crown a universal winner among the three optimisers but to map which regimes preserve or lose rare-class performance. The design separates three axes the FL literature often conflates. *Statistical heterogeneity* controls which clients own the rare-class signal — from IID, through Dirichlet label skew (Hsu et al., 2019), to class-disjoint clients — and is fixed before training by the data partition; it determines *where* rare-class information resides. *System heterogeneity* controls whether that signal reaches aggregation: a straggler completes fewer local steps, and its partial update may be dropped or accepted, or it may take smaller steps under a lower learning rate. *Optimiser and protocol choices* then determine the global response. The two heterogeneities interact most sharply when the client carrying the rare classes is also the one that is slow or under-weighted.

This motivates the concept that organises the thesis, *rare-client signal flow*: the per-round contribution that the rare-class-holding client makes to the global update. A rare class is learned only to the extent that the information in that client's update d_i reaches and persists in w^{t+1}, so the organising hypothesis is that rare-class performance is preserved precisely when rare-client signal continues to reach the global model. The matched two-client partitions L4 (class-disjoint) and L1 (quantity-only control) isolate this (§2.4). Performance is measured throughout by test macro-F1 at the best-validation checkpoint (§2.2).

The optimiser set is deliberately limited rather than exhaustive, and the four-condition decomposition lacks a FedAvg + γ-inexact cell, so its rescue attribution is to the *FedProx + γ-inexact protocol*, not to γ-inexact acceptance alone.

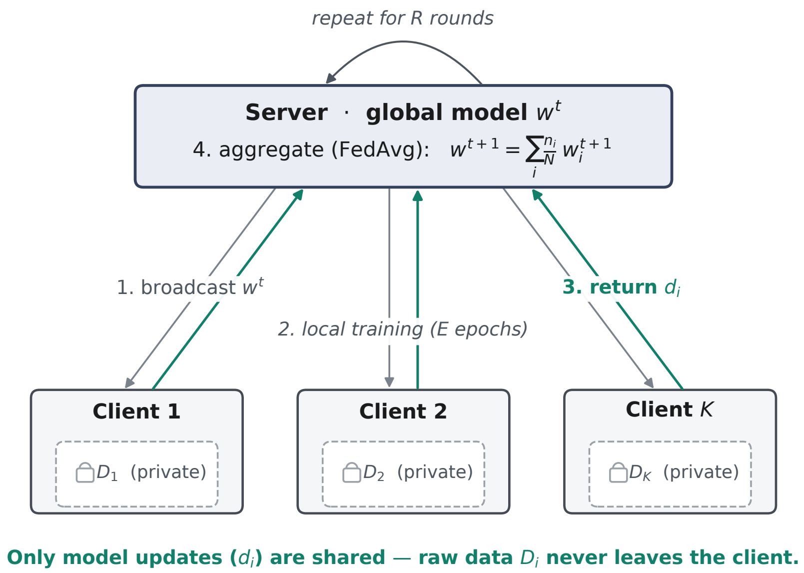

Figure 2.1: Federated learning loop (conceptual; *simulated* on a single machine in this thesis). Each round the server broadcasts the global model w^t, clients train locally on private data D_i and return model updates d_i (not raw data), and the server aggregates them into w^{t+1}; the loop repeats for R rounds.

2.2 Dataset, metrics, and reporting

All experiments use DermaMNIST from MedMNIST v2 (Yang et al., 2023), a seven-class dermatology benchmark derived from the HAM10000 dataset of dermatoscopic skin-lesion images (Tschandl et al., 2018). The task is single-image classification at 28×28 RGB resolution. Splits follow the canonical MedMNIST v2 release with 7007 training, 1003 validation, and 2005 test images, permuted to channel-first float, scaled to $[0, 1]$, and normalised with ImageNet channel statistics.

The benchmark is severely class-imbalanced (Figure 2.2, Table 2.1): mel-nevi dominates the training set, while dermatofibroma and vascular lesions have so little support that failing to learn them incurs a large macro-F1 penalty at almost unchanged accuracy. Melanoma is mid-support but is kept in the canonical rare-class group RARE_IDX $= \{\text{dermato}, \text{melanoma}, \text{vascular}\}$, since missing a melanoma is the highest-priority failure. Mel-nevi acts as the *majority attractor*: under failing protocols, rare-class images are routed into it, so accuracy stays high while macro-F1 collapses. Partitions (§2.4) apply to the training set only; validation and test sets stay global and class-stratified, so any test-set rare-class collapse is not a partition artefact.

Table 2.1: DermaMNIST class supports and roles in this thesis. Train prevalence and test support are computed directly from the MedMNIST v2 release; the train support totals 7007 and the test support totals 2005. The rare-class group RARE_IDX = $\{\text{dermato}, \text{melanoma}, \text{vascular}\}$ defines the rare-class F1 used throughout Chapter 3.

Class	Train prevalence	Test support	Role in thesis
mel-nevi (melanocytic nevi)	66.9 %	1341	Majority attractor; collapse-mode prediction class
melanoma	11.1 %	223	Mid-support, high clinical priority; in rare-class group
dermatofibroma	1.1 %	23	Smallest class; in rare-class group
vascular lesions	1.4 %	29	Second smallest; in rare-class group
remaining (akiec, bcc, b-kerat)	19.4 %	389 total	Per-class reporting only; no special role

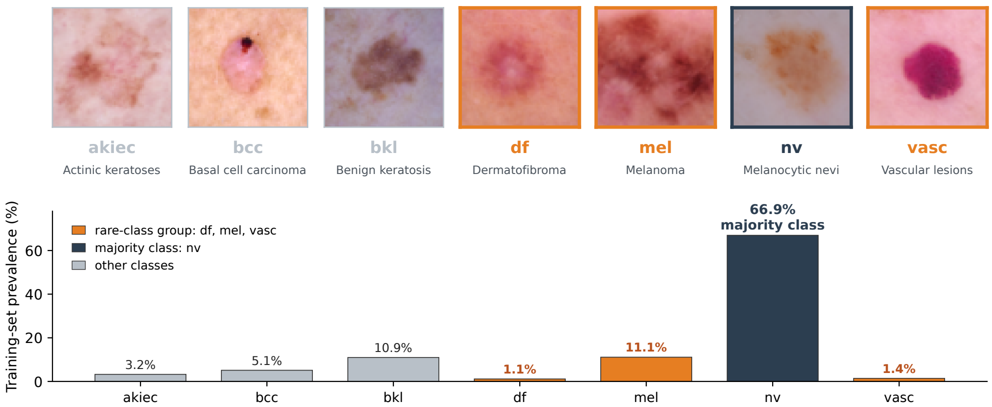

Figure 2.2: DermaMNIST: one example image per class (top) and training-set class prevalence (bottom). Melanocytic nevi (nv) dominate at 66.9 %, while the rare classes used throughout — dermatofibroma (1.1 %), vascular (1.4 %), and melanoma (11.1 %; orange) — are a small fraction of the data. This imbalance is why raw accuracy is misleading and the thesis ranks optimisers by macro-F1.

The primary metric throughout is *test macro-F1 at the best-validation checkpoint*, computed on the global test set: macro-F1 weights all seven classes equally and is sensitive to rare-class collapse, whereas raw accuracy saturates at the $\approx 66.9\,\%$ mel-nevi floor. Secondary diagnostics are per-class F1; per-class recall (the confusion-matrix diagonal $P(\hat{y} = c \mid y = c)$); rare-class F1 over RARE_IDX; rare-to-mel-nevi misrouting, the mean over rare classes of $P(\hat{y} = \text{nv} \mid y = c)$; convergence curves; and per-client update norms (diagnostic only). Balanced accuracy is stored only as a sanity check, never used to rank optimisers. Validation macro-F1 selects the best checkpoint round-by-round; test metrics are evaluated once at that checkpoint. Evaluation uses argmax predictions. A *protocol flip* is a sign reversal of $\Delta = \text{FedProx} - \text{FedAvg}$ between the best-validation, final-round, and last-K summaries, reported as a robustness check.

Three seed tiers are reported distinctly (Table 2.2); sample SDs use the unbiased $(N - 1)$ denominator. The *node-pinned L4* configuration is a variance control: it re-runs L4 cells on a single compute node with matched seeds to isolate cross-node CUDA non-determinism from any optimiser effect.

Table 2.2: Seed-tier reporting protocol. Wins is the number of paired within-seed comparisons in which FedProx exceeds FedAvg. Wilcoxon refers to the paired signed-rank test (Demšar, 2006), used only where $n = 10$ provides sufficient power.

Tier	Statistical reporting	Use case
$n = 10$	Mean ± sample SD; paired Δ; wins; paired Wilcoxon signed-rank where meaningful	Headline statistical-heterogeneity, μ-sweep, IID baseline, system-het 7-client experiments
$n = 3$	Mean ± sample SD; directional only; no significance language	Mechanism/asymmetry probes: four-condition, perfect-storm, LR asymmetry, weighted-CE, asymmetric-μ
$n = 1$	Labelled *pilot* (single seed)	Heterogeneity ladder L0–L4

2.3 Simulation and model

All experiments are controlled federated-learning simulations. Each run executes the loop of Figure 2.1 for $R = 150$ rounds with $E = 20$ local epochs of mini-batch SGD per client, using the same global, class-stratified validation and test sets throughout — only the client training partition varies. Two federation sizes are used: the **7-client** federation carries the headline 10-seed benchmarks and the random-straggler system-het comparisons, while the **2-client** federation carries the controlled mechanism probes that need a clean rare-client vs dominant-client interaction (no cross-scale claim is made between the two). Two training regimes are used: **Regime A** (7-client default: SGD momentum 0.9, batch 32, lr = 0.01, weight decay 0) and **Regime B** (Li-style plain SGD for the 2-client probes: momentum 0, batch 10, same lr/wd/E/R); the four-condition decomposition additionally fixes Client 1's $E_{\text{straggler}} = 5$. Each experiment's regime is stated in its Results section.

The model is a small CNN used as the fixed benchmark vehicle, so that optimiser differences are attributable to the federated method rather than the architecture: four convolutional blocks of widths $32 \to 64 \to 128 \to 256$ (each a 3×3 convolution with GroupNorm (Wu and He, 2018) and ReLU; max-pooling between the first three, adaptive average pooling after the fourth), then a $256 \to 128$ dense layer with dropout 0.2 and a seven-class head. GroupNorm is used rather than BatchNorm because BN's per-batch statistics diverge across heterogeneous clients, a documented non-IID FL pathology (Hsieh et al., 2020; Li et al., 2021). The local objective is unweighted cross-entropy unless §2.4.3 overrides, with best-validation checkpointing. Models were implemented in PyTorch (Paszke et al., 2019); macro-F1, per-class F1, and confusion matrices were computed with scikit-learn (Pedregosa et al., 2011), and statistical diagnostics with SciPy (Virtanen et al., 2020).

Two runtimes are used: a *pure-PyTorch reference loop* and the *Flower simulation* (framework 1.23 (Beutel et al., 2020)), which carries every experiment and all reported headline numbers. Where the two disagree on magnitude (the engineered partition, §3.1), Flower is the central runtime and the pure-PyTorch estimate is direction-only corroboration. Full implementation provenance, saved-artefact formats, and reproducibility guards are given in Appendix A.2.

2.4 Partitions, optimisers, and protocols

Partition design is the primary lever for statistical heterogeneity, determining whether each client sees a near-global class mix or rare-class signal is concentrated on a few clients — or, in the strongest probes, on a single client whose contribution can then be made to fail or succeed by the system-heterogeneity protocol (Table 2.3). The *IID* partition is a negative control: every client receives a representative sample, so a substantial FedProx advantage here would not support the heterogeneity-correction interpretation. The *engineered balanced-paired* 7-client partition is a controlled middle ground in which every minority class is concentrated on exactly two clients (rare-class signal unevenly shared) but none is isolated to a single client. It is the substrate for the headline FedAvg/FedProx comparison and the μ-sweep (Figure 2.3); it was deliberately *"designed to maximise FedProx's advantage on the per-class drift mechanism"*, so the small Flower-paired Δ in §3.1 should be read against that FedProx-favouring framing rather than as a general effect estimate.

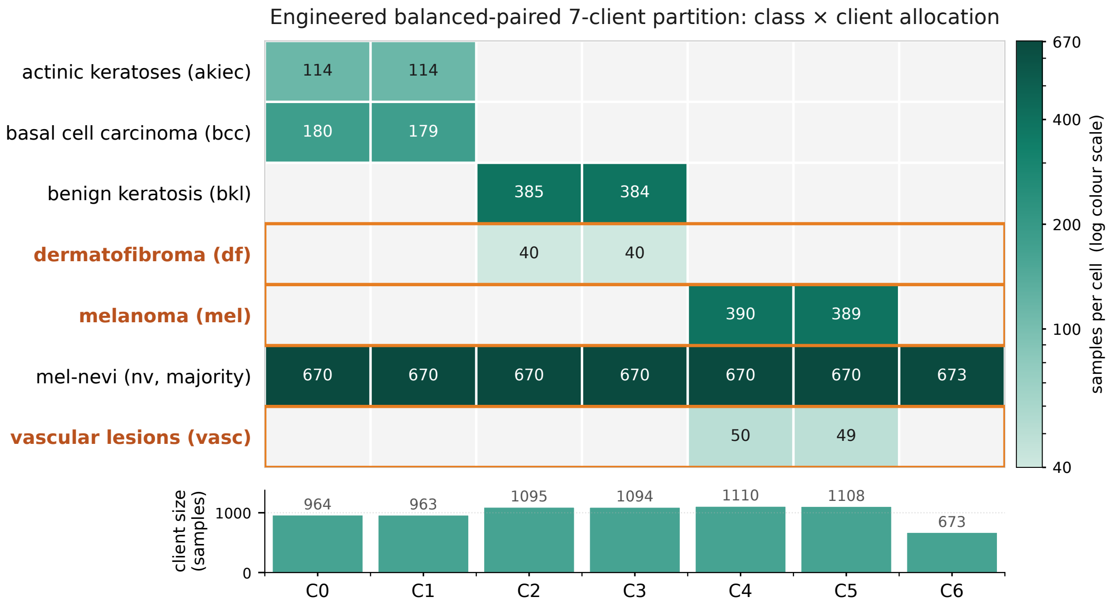

Figure 2.3: Engineered balanced-paired 7-client partition: training-sample counts per (class, client) cell. Rare classes (df, mel, vasc; outlined in red) sit on exactly two clients each; mel-nevi (majority) is broadly distributed across all seven clients in ≈ 670-image shares plus a 673-image generalist on C6. This creates controlled statistical heterogeneity — rare-class signal is unevenly shared — without reducing any rare class to a singleton specialist case. The bottom bar gives the total training-sample count per client.

The two-client **L1** and **L4** partitions are matched mechanism controls (Figure 2.4): both use the same 86/14 split and protocol family and differ in one variable — whether Client 1 carries class information Client 0 lacks. *L4* is class-disjoint: Client 1 uniquely holds melanoma, dermatofibroma, and vascular lesions (JS divergence 1.0), and is the substrate for the four-condition decomposition, perfect-storm, asymmetric-LR, and loss-side experiments. *L1* keeps the same quantity skew but stratifies every class across both clients (JS divergence $\approx 5\times10^{-6}$), so it tests whether γ-inexact acceptance helps generically, while L4 tests whether it helps specifically when the under-trained client holds unique rare-class signal.

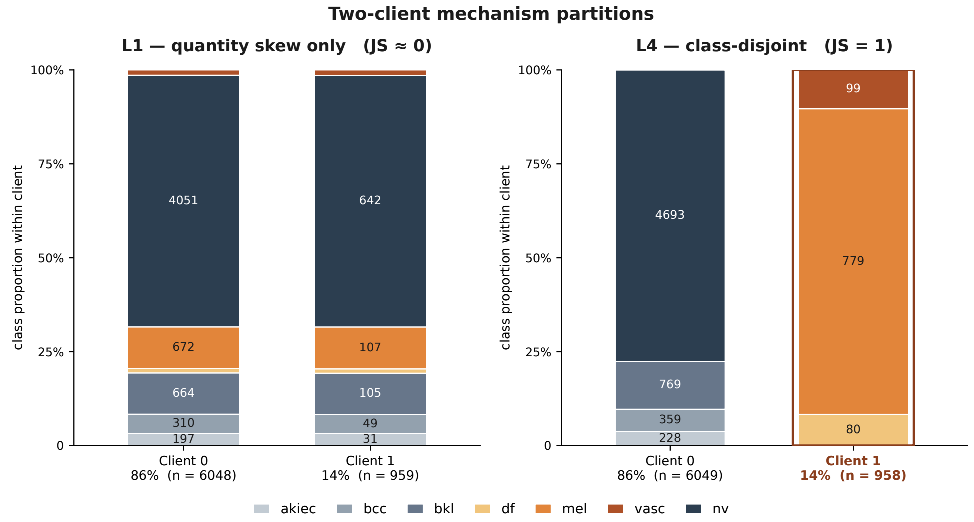

Figure 2.4: Two-client mechanism partitions. **L1** keeps the (86/14) quantity split but gives both clients near-global class proportions. **L4** is class-disjoint: Client 0 holds the common classes and Client 1 uniquely holds melanoma, dermatofibroma, and vascular lesions. The contrast isolates whether the under-trained client carries unique rare-class signal.

Table 2.3: Client-partition designs used in the chapter. The engineered balanced-paired partition and the L1/L4 pair are load-bearing; the remaining partitions test robustness or provide pilot context.

Partition	Federation	Statistical-het type	Purpose	Results ğ
IID	7-client	none	Negative control for FedProx under no statistical heterogeneity	ğ3.1
Engineered balanced-paired	7-client	paired minority ownership	Main FedAvg/FedProx comparison and μ-sweep	ğ3.1, ğ3.2
Dirichlet $\alpha = 0.1$	7-client	stochastic label skew	Test whether the engineered result generalises	ğ3.1
Specialist 1-of-7	7-client	singleton minority ownership	Test stronger class specialisation without changing client sizes	ğ3.1
Ladder L0–L4	2-client	escalating label skew	Single-seed pilot	ğ3.1 (pilot)
L1	2-client	quantity-only	Negative control for the γ-inexact mechanism	ğ3.3
L4	2-client	class-disjoint	Mechanism substrate for straggler, LR-asymmetry, and loss experiments	ğ3.3, ğ3.4, ğ3.5

2.4.1 Optimisers

Optimiser choice and straggler-handling protocol are treated as *independent* axes: the optimisers below define how each client's local update is produced and combined, while drop versus γ-inexact handling separately determines whether a straggler's partial update enters aggregation. This separation is what lets the four-condition decomposition attribute the rare-class-rescue effect to one mechanism rather than the other (Table 2.4). **FedAvg** (McMahan et al., 2017) is the baseline: clients run E epochs of local SGD and the server takes the sample-weighted mean, with no drift control or work normalisation, so on L4 the rare-class client ($\approx 14\,\%$ of samples) is structurally under-weighted. **FedProx** (Li et al., 2020) keeps that aggregation rule

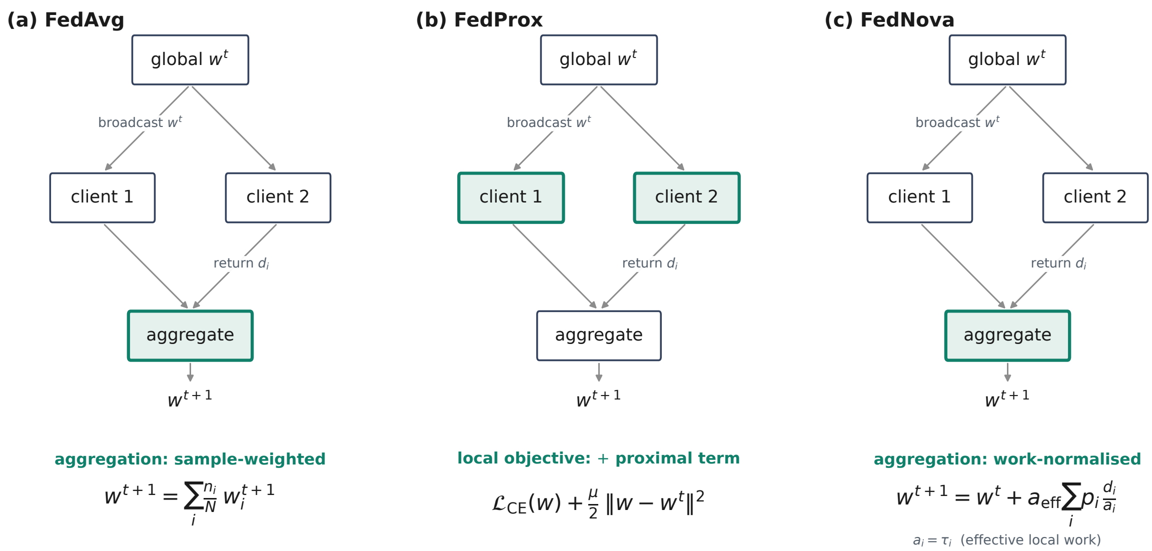

Figure 2.5: Minimal schematic comparison of the three federated optimisers. All three share the same structure — broadcast the global model, train on each client, aggregate — and differ only in the highlighted mechanism. FedAvg aggregates locally trained client models by sample-weighted averaging; FedProx keeps that aggregation but adds a proximal penalty to the *local* objective, discouraging drift from the current global model; FedNova normalises client updates by effective local work before aggregation. The diagram is conceptual; Chapter 3 tests when these mechanisms preserve or lose rare-client signal under statistical and system heterogeneity.

but augments each client's local objective with a proximal penalty

$$\mathcal{L}_{\text{prox}}(w; w_{\text{global}}) \;=\; \mathcal{L}_{\text{CE}}(w) \;+\; \tfrac{\mu}{2}\, \|w - w_{\text{global}}\|^2,$$

where w_{global} is the round-start model and μ the proximal coefficient (default $\mu = 0.01$ outside the μ-sweep of §3.2). By construction $\mu = 0$ makes the update bit-identical to FedAvg, an equivalence verified numerically by a regression test, so any later FedProx advantage cannot be a coding artefact. **FedNova** (Wang et al., 2020) keeps FedAvg-style solvers but normalises each client's update by its effective local work τ_i via the momentum-aware factor

$$a_i = \frac{\tau_i(1-m) - m\,(1-m^{\tau_i})}{(1-m)^2},$$

which reduces to $a_i = \tau_i$ when $m = 0$ (as under Regime B). The server aggregates d_i/a_i and scales the global step by $a_{\text{eff}} = \sum_i p_i a_i$; both FedNova's asymmetry regimes lie outside the stable-τ assumption of the original analysis. Figure 2.5 contrasts these three mechanisms against the shared federated-learning loop.

Table 2.4: Federated optimisation methods used in this thesis. The "mechanism" column names the operative change relative to FedAvg; "main knob" is the hyperparameter the experimenter actually moves; "expected vulnerability" summarises the regime in which the method is predicted to underperform, and the right-most column locates the Results section in which that prediction is tested.

Method	Mechanism	Main knob	Expected vulnerability	Results ğ
FedAvg	Sample-weighted aggregation; no drift control	none beyond local SGD	Rare-class client structurally under-weighted under quantity skew	ğ3.1, ğ3.3
FedProx	FedAvg plus local proximal penalty $\frac{\mu}{2}\|w - w_{\text{global}}\|^2$	μ (default 0.01)	μ inverted-U: too small $\approx$ FedAvg; too large overconstrains rare-class learning	ğ3.2, ğ3.3
FedNova	Sample-weighted aggregation of τ_i-normalised local directions d_i/a_i	client momentum m, local-step count τ_i	Fails under random per-round τ (outside the stable-τ assumption of (Wang et al., 2020))	ğ3.4

2.4.2 Partial-update protocols and stress regimes

The L1/L4 partitions and the drop and straggler protocols below are stylised stress tests: they push statistical and system heterogeneity to extremes to expose *when* rare-client signal survives or disappears. Such extremes are not purely hypothetical — a real site may genuinely lack whole lesion classes — and the claims here concern method behaviour at these boundaries.

System heterogeneity is simulated by reducing certain clients' local work in a round (computational asymmetry, not wall-clock latency) under one of three schedules: uniform (all clients run the full E); fixed-straggler (named clients run $E_{\text{straggler}} < E_{\max}$); or random-straggler (a fraction per round run a random $E \in [1, E_{\max} - 1]$). When a client returns a partial update the server either *drops* it (excluded from that round's aggregation) or accepts it under *γ-inexact* handling (Figure 2.6). **Drop versus γ-inexact is a protocol choice independent of the optimiser:** either policy pairs with either FedAvg or FedProx, and this independence is what the four-condition decomposition exploits.

The *four-condition decomposition* runs on L4 under Regime B with the fixed-straggler schedule ($E_{\max} = 20$, $E_{\text{straggler}} = 5$ on Client 1, the rare-class holder), $n = 3$ matched seeds per cell. Holding the partition and schedule fixed isolates the drop-vs-accept axis from any per-round variability, so the only quantities that change between the four cells are the optimiser (FedAvg vs FedProx) and the partial-update protocol (drop vs γ-inexact); the full cell-by-cell design grid is tabulated in Appendix A.3 (Table A.2). The design deliberately omits a FedAvg + γ-inexact cell, so attributing the rare-class-rescue effect to the γ-inexact protocol (rather than to the proximal term alone) is a claim about the *FedProx + γ-inexact* protocol, not about γ-inexact acceptance in isolation.

Three further system-asymmetry regimes build on this. The *perfect-storm* stress keeps L4/Regime B but raises straggling to a random fraction 0.9, comparing FedAvg+drop with FedProx+γ-inexact at $\mu \in \{1.0, 0.01\}$ ($n = 3$); because its schedule and arms differ from the four-condition design it is reported as a stress demonstration of the same ordering. The

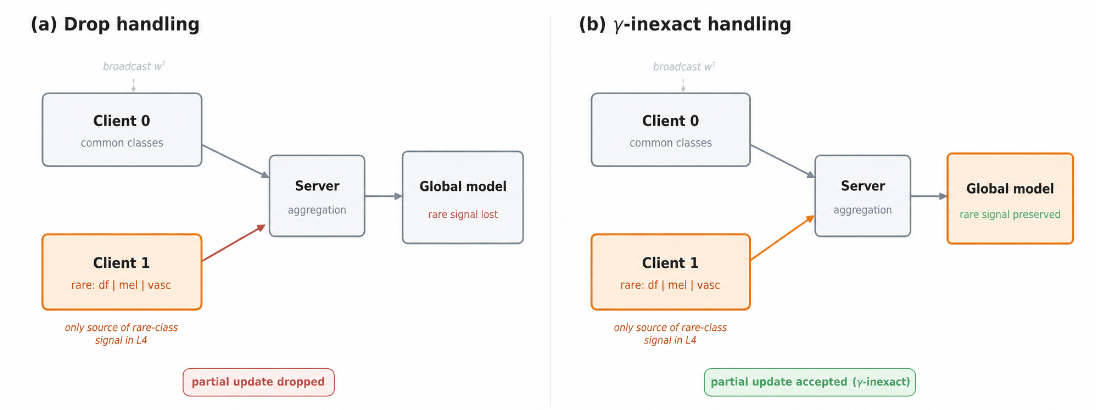

Figure 2.6: Rare-client signal flow under drop and γ-inexact partial-update handling in the L4 partition. Client 1 uniquely holds melanoma, dermatofibroma, and vascular examples. Under **(a)** drop handling its partial update is discarded, so rare-class signal is lost; under **(b)** γ-inexact handling the partial update enters aggregation and the signal is preserved. The schematic is conceptual; the evidence in §3.3 concerns the FedProx + γ-inexact protocol in this design.

learning-rate-asymmetry envelope keeps Client 0 at $\text{lr} = 0.01$ and varies Client 1 across six ratios from 1:1 to 50:1 ($\text{lr}_1 \in \{0.01, 0.005, 0.002, 0.001, 0.0005, 0.0002\}$), weakening the rare client's update magnitude *without dropping it* and so separating the "silent client" failure mode from the "smaller-step" one. The *random-τ FedNova stress* runs on the 7-client engineered partition under Regime A with random-straggler fraction 0.5, $n = 10$ — not an L4 experiment — deliberately making τ_i a per-round random variable that violates the stable-τ assumption of Wang et al. (2020).

2.4.3 Loss-side interventions

The loss-side comparators test a falsification hypothesis: if FedProx's advantage is really a disguised class-imbalance correction, a direct loss-level correction should match or exceed it on the same partition. Three losses are compared on the L4 substrate: plain cross-entropy (the unweighted baseline); weighted CE with inverse-frequency weights $w_c = (1/n_c) \cdot C / \sum_j (1/n_j)$ normalised so $\sum_c w_c = C$; and focal loss (Lin et al., 2017) ($\gamma = 2$, inverse-frequency α_c). The focal comparator uses this repository's class-balanced variant, in which the modulating factor is computed from the α-weighted cross-entropy rather than from unweighted log-probabilities, so it is an empirical loss-side comparison. Weighted CE intervenes in per-class loss during local training while FedProx intervenes in aggregation, so if FedProx were class re-weighting in disguise, weighted CE would close the gap on its own. This comparator set is run only on L4.

2.5 Experiment families

Table 2.5 maps the eleven experiment families to the Results sections they support; the Results chapter follows it as a single argument rather than a list of detached experiments. L4 is the controlled-mechanism substrate (five rows); L1 appears once but is the load-bearing negative control for the γ-inexact rescue mechanism (§3.3).

Table 2.5: Experiment families and the Results sections they support. The table bridges the design choices defined in ğ2.1–ğ2.4.3 and Results (Chapter 3). Each Results section examines one row or a group of rows; none is presented as a detached standalone experiment.

Experiment family	Federation / partition	Algorithms / protocols	What it tests	Results ğ
IID control	7-client IID	FedAvg vs FedProx, $n = 10$	Falsify FedProx-helps-without-heterogeneity	ğ3.1
Engineered statistical-het	7-client balanced-paired	FedAvg vs FedProx, both runtimes, $n = 10$	Headline Δ under engineered label skew	ğ3.1
Robustness partitions	7-client Dirichlet / specialist	FedAvg vs FedProx, $n = 10$	Test engineered direction outside its framing	ğ3.1
Node-pinned L4 control	2-client L4, single compute node	FedAvg vs FedProx, $n = 3$ matched	Isolate cross-node CUDA non-determinism	ğ3.1
μ sensitivity	7-client balanced-paired	FedProx at $\mu \in \{0, 0.001, 0.01, 0.1, 1.0\}$, $n = 10$	Locate proximal-coefficient peak; show large-μ harm	ğ3.2
Li-style four-condition	2-client L4 and L1 (control)	FedAvg/FedProx $\times$ drop/γ-inex, $n = 3$	Isolate proximal-only from γ-inex contribution	ğ3.3
Perfect-storm	2-client L4, batch 10, momentum 0, 90% random stragglers	FedAvg+drop vs FedProx+γ-inex ($\mu \in \{1.0, 0.01\}$), $n = 3$	Stress demonstration of the four-condition ordering	ğ3.3
LR asymmetry	2-client L4, 6 ratios 1:1 to 50:1	FedAvg, FedProx, FedNova, $n = 3$ per ratio	FedNova robustness under deterministic LR skew	ğ3.4
FedNova random-τ	7-client balanced-paired, $f_{\text{straggle}} = 0.5$	FedNova vs system_het_random FA/FP, $n = 10$	Random-τ stress outside Wang 2020's stable-τ assumption	ğ3.4
Loss-side correction	2-client L4	FedAvg/FedProx $\times$ CE / weighted CE / focal, $n = 3$	Test whether FedProx is a disguised class-imbalance corrector	ğ3.5
Mechanism diagnostics	cross-protocol pool of L4 runs	update-norm trajectories, protocol-flip audit	Drift-control mechanism evidence; reporting-protocol robustness	ğ3.6

Seed counts vary across experiments because compute availability varied; each experiment's seed count is reported transparently in the table above.

3 Results

Every configuration is evaluated by test macro-F1 at the best-validation checkpoint (§2.2): on DermaMNIST, where melanocytic nevi account for 66.9 % of the data, accuracy saturates at the majority floor and hides rare-class failure, whereas macro-F1 weights all seven classes equally. Three seed tiers are used throughout: $n = 10$ paired seeds for headline comparisons, $n = 3$ as directional evidence for mechanism probes, and single-seed pilots labelled as such, with no significance test applied to the $n = 3$ directional probes; the full metric hierarchy and reporting protocol are tabulated in Appendix A.3 (Table A.1).

The results proceed from weakest to strongest stress: statistical heterogeneity, the proximal coefficient, straggler asymmetry, learning-rate asymmetry and FedNova, rare-class collapse, and mechanism checks.

3.1 Weak separation under statistical heterogeneity

This first result asks whether statistical heterogeneity alone is enough to separate FedAvg and FedProx; the IID split serves as the negative control. Under an IID partition (Flower, $n = 10$ paired seeds), FedAvg and FedProx differ by $\Delta = -0.007$ macro-F1 (FA 0.585, FP 0.578, FN 0.577) — within seed noise. We treat this as a negative control: statistical heterogeneity is a necessary condition for any FedAvg–FedProx gap on this benchmark.

On the engineered balanced-paired seven-client partition, three estimates ($n = 10$ paired seeds each; Table 3.1) agree on sign and bracket the magnitude at 0.01–0.03: the central headline is the Flower paired $\Delta = +0.007$ (7/10 wins, full provenance), with the μ-sweep peak (+0.013) and the pure-PyTorch reference (+0.027, 9/10 wins) as cross-runtime corroboration.

The pure-PyTorch +0.027 is concentrated in two minority classes — melanoma and actinic keratoses carry 61% and 36% of the macro-F1 advantage — while the other five classes, including the invariant mel-nevi, are flat (Figure 3.1). The gain is concentrated in the rebalancing of two minority classes.

Three further partitions replicate the small-magnitude finding — Dirichlet $\alpha = 0.1$, specialist 1-of-7 ($n = 10$ each), and the multi-seed node-pinned L4 control ($n = 3$), all with small positive Δ (Table 3.1). The single-seed heterogeneity-ladder pilot at L4 gives $\Delta = -0.030$, but that deficit is one rare-class (vascular) outcome on one seed and the $n = 3$ node-pinned promotion averages it out. (Dirichlet aggregates use regenerated raw-JSON summaries.)

On the engineered partition, FedProx reduces test-macro-F1 SD from 0.025 to 0.014 (−45 %), reaches validation macro-F1 = 0.40 in 9.8 vs 12.7 rounds (1.3× speed-up), and shows a paired mid-training peak $\Delta = +0.094$ at round 50 that closes to +0.007 by round 150. These trajectory characteristics are documented for the engineered partition only.

Statistical heterogeneity alone produces small, runtime-sensitive separation (Table 3.1). The next question is whether the proximal coefficient μ itself explains when FedProx helps.

Table 3.1: Statistical heterogeneity and runtime sensitivity: test macro-F1 (mean ± sample SD where shown) for FedAvg vs FedProx across six settings — IID, the engineered partition through both runtimes plus the μ-peak, Dirichlet $\alpha = 0.1$, specialist 1-of-7, and the node-pinned L4 control. IID, Dirichlet, and specialist values are from regenerated raw-JSON aggregates.

Setting	Runtime	Clients	n	FedAvg	FedProx	Δ	Note
IID	Flower	7	10	0.585 ± 0.020	0.578 ± 0.020	-0.007	negative control
Engineered (pure)	pure-PyTorch	7	10	0.481 ± 0.025	0.508 ± 0.014	$+0.027$	9/10 wins; partial provenance
Engineered (Flower)	Flower	7	10	0.496	0.503	$+0.007$	7/10 wins; **central**
Dirichlet $\alpha = 0.1$	Flower	7	10	0.480 ± 0.036	0.488 ± 0.034	$+0.008$	regenerated raw-JSON
Specialist 1-of-7	Flower	7	10	0.437 ± 0.024	0.445 ± 0.017	$+0.008$	—
Node-pinned L4	Flower	2	3	0.492 ± 0.019	0.497 ± 0.022	$+0.005$	multi-seed L4 control

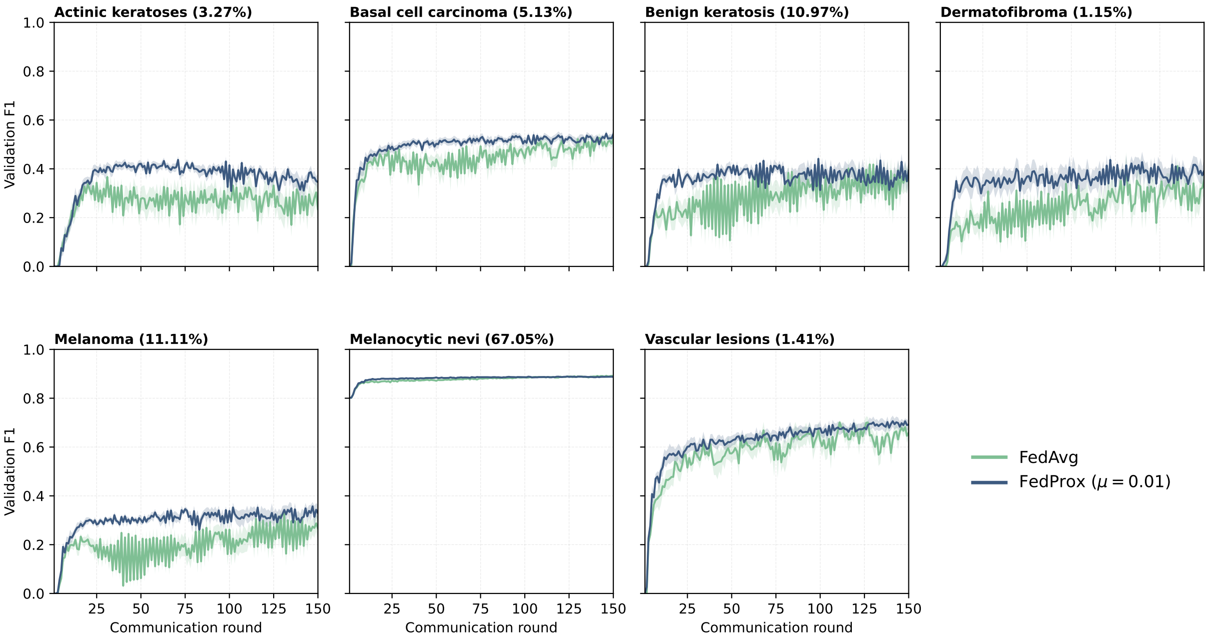

Figure 3.1: Per-class validation F1 trajectories on the engineered partition from the *pure-PyTorch* headline runs, $n = 10$ paired seeds (FedAvg vs FedProx, mean ±1 SEM bands). By mid-training the melanoma and actinic-keratoses panels show non-overlapping FedAvg/FedProx bands — the two minority classes that carry the engineered FedProx advantage — while the other five classes overlap throughout 150 rounds, including the dominant mel-nevi panel, which stays invariant near its prevalence saturation.

3.2 Proximal coefficient: a narrow useful range

The next question is whether the proximal coefficient μ — the one knob distinguishing FedProx from FedAvg — governs the advantage, and whether larger μ helps under heterogeneity. A five-cell sweep ($\mu \in \{0.001, 0.01, 0.1, 1.0\}$ against the $\mu = 0$ FedAvg baseline; engineered partition, $n = 10$ paired seeds) traces a clear inverted-U (Table 3.2, Figure 3.2): peak near $\mu = 0.01$, below baseline by $\mu = 0.1$, catastrophic at $\mu = 1.0$.

The peak is at $\mu = 0.01$ ($\Delta = +0.013$, 6/10 wins), essentially tied with $\mu = 0.001$; by $\mu = 0.1$ FedProx falls below baseline, and at $\mu = 1.0$ it collapses ($\Delta = -0.074$, 0/10 wins; Table 3.2). The 0.09 macro-F1 spread between best and worst μ is an order of magnitude larger than any partition-level Δ in § 3.1.

The μ-sweep peak $\Delta = +0.013$ exceeds the § 3.1 Flower paired $\Delta = +0.007$ because the sweep selects best-of-five on the same seeds; the two estimates bracket the same small effect at $+0.005$ to $+0.015$.

The practical region of useful μ on this benchmark is a single decade around 0.01. The expectation that larger μ should be preferred under stronger statistical heterogeneity is not supported here — a benchmark observation specific to this setting.

The $\mu = 1.0$ deficit concentrates in the rare and minority classes while the dominant mel-nevi is essentially invariant (Table A.3) — the same rare-class shape as the dropped-straggler and asymmetric-LR failures (§ 3.5), via a different mechanism (over-regularisation rather than under-aggregation). Because μ alone is sensitive and harmful when large, the much larger FedProx gains under straggler protocols (§ 3.3) cannot be attributed to the proximal coefficient itself.

Table 3.2: μ sensitivity on the engineered partition, $n = 10$ paired seeds per cell. "Wins" is paired comparisons where FedProx exceeds FedAvg on the matched seed.

μ	macro-F1 (mean ± SD)	Δ vs FedAvg	Wins / 10
0	0.493 ± 0.020	—	—
0.001	0.504 ± 0.021	$+0.011$	6
0.01	$\mathbf{0.506 \pm 0.017}$	$\mathbf{+0.013}$	6
0.1	0.479 ± 0.017	-0.014	2
1.0	0.419 ± 0.013	-0.074	0

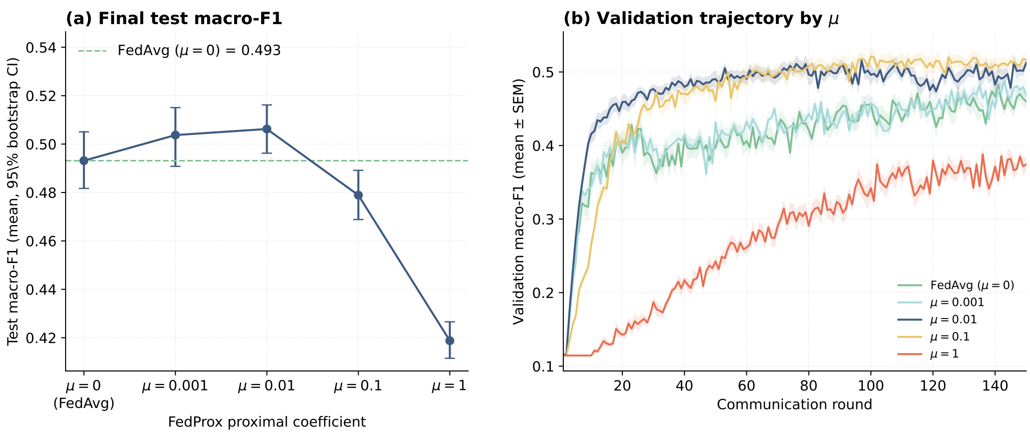

Figure 3.2: Inverted-U μ sensitivity curve on the engineered partition ($n = 10$ per cell). x-axis: μ on $\log_{10}$ scale across $\{0.001, 0.01, 0.1, 1.0\}$, FedAvg baseline as dashed reference line at 0.493. y-axis: test macro-F1 with ± 1 SD whiskers.

3.3 Straggler asymmetry: γ-inexact handling, not the proximal term

This section presents the central result of the thesis. Every preceding section has shown the FedAvg–FedProx gap to be small; here it becomes large, and a controlled four-condition decomposition attributes that gap to its true cause — not the proximal regulariser FedProx is named for, but the γ-inexact partial-update handling it is conventionally paired with. This is what turns the thesis from an optimiser comparison into a regime map.

Before decomposing the mechanism, we first checked whether stragglers alone create a FedProx advantage. They do not: in a 7-client IID random-straggler control FedProx is slightly worse than FedAvg ($\Delta = -0.013$ macro-F1, $n = 10$), but replacing the IID split with a heterogeneous one under the same straggler protocol reverses the sign ($\Delta = +0.017$, $n = 10$). The advantage appears only when slow or partially updated clients also carry non-redundant class information; the next experiment isolates that mechanism in a two-client L4 decomposition with the rare-class client as the fixed straggler.

We next isolate the mechanism in a two-client L4 four-condition experiment, with $n = 3$ matched seeds (Table 3.3). In this setting, Client 1 contains the rare-class signal and is made the fixed straggler. The four cells separate two effects that are otherwise bundled together in FedProx-style straggler experiments: the proximal term and the decision to accept or drop a straggler's partial update.

Table 3.3: Li-style four-condition decomposition at L4 (two-client; $n = 3$ matched seeds; mean $\pm$ sample SD across seeds).

Condition	Test macro-F1	Rare-class F1
FedAvg, no straggler	0.485 ± 0.028	0.316 ± 0.060
FedAvg + drop	0.364 ± 0.009	0.000 ± 0.000
FedProx + drop (control)	0.368 ± 0.005	0.000 ± 0.000
FedProx + γ-inexact	$\mathbf{0.479 \pm 0.010}$	$\mathbf{0.291 \pm 0.018}$

Decomposition: proximal-only $= 0.368 - 0.364 = +0.004$; γ-inexact $= 0.479 - 0.368 = +0.111$; total $= +0.115$. Rare-F1 is the mean over RARE_IDX $= \{\text{dermato}, \text{melanoma}, \text{vascular}\}$: both drop-handling cells sit at zero and only FedProx+γ-inexact recovers it.

Dropping the straggler removes class information rather than merely slowing training: FedAvg falls from 0.485 to 0.364. Adding the proximal term while still dropping changes almost nothing (+0.004), whereas accepting the partial update recovers nearly the full baseline (+0.111). About 96% of the +0.115 gain is therefore γ-inexact acceptance, not the proximal term. Because the design omits a FedAvg+γ-inexact cell, this is evidence for the FedProx+γ-inexact protocol specifically; the standalone effect of γ-inexact acceptance is left for future isolation.

Figure 3.3 (A) shows the same result over training: the FedAvg+drop and FedProx+drop trajectories overlap for all 150 rounds, while FedProx+γ-inexact separates from both around round 30 and stabilises near 0.48.

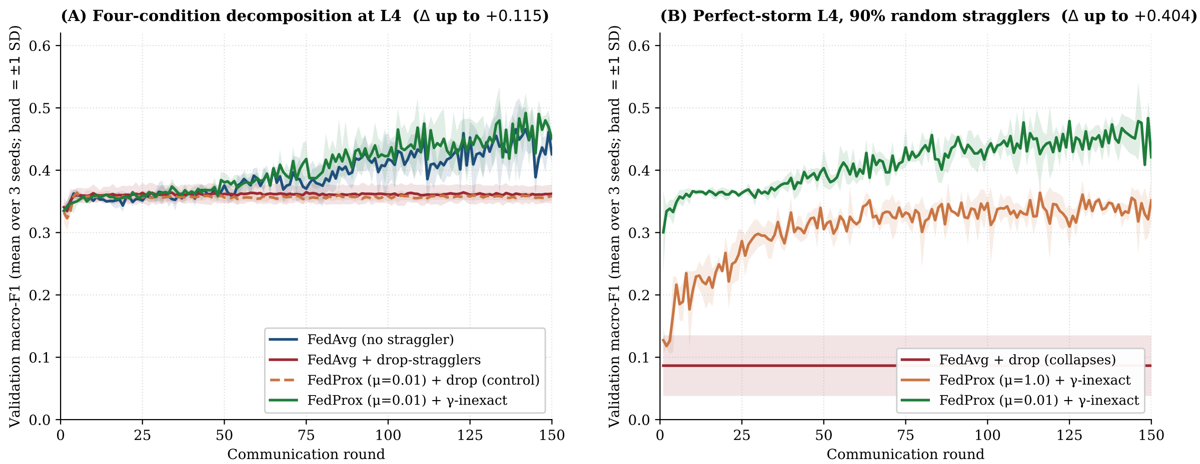

Figure 3.3: Validation macro-F1 trajectories at the two settings with the largest FedAvg vs FedProx differences (mean over three seeds; shaded bands are ±1 sample SD). (A) The four-condition decomposition at L4: the FedAvg+drop and FedProx+drop trajectories overlap throughout the 150 rounds; only FedProx+γ-inexact separates and recovers towards the no-straggler baseline (Δ up to +0.115). (B) Perfect-storm L4 with 90% random stragglers: FedAvg+drop flatlines at its round-1 validation value, so the selected checkpoint is effectively the untrained model, while FedProx+γ-inexact climbs to 0.491 at $\mu = 0.01$ (Δ up to +0.404 — the largest FedProx–FedAvg gap measured in the thesis).

Per-class trajectories (Figure 3.4) localise the gain to the rare classes: mel-nevi stays near 0.89 under all four conditions, while dermatofibroma, melanoma, and vascular collapse under both drop cells and recover only under FedProx+γ-inexact — the mechanism is rare-class learning, not a broad improvement.

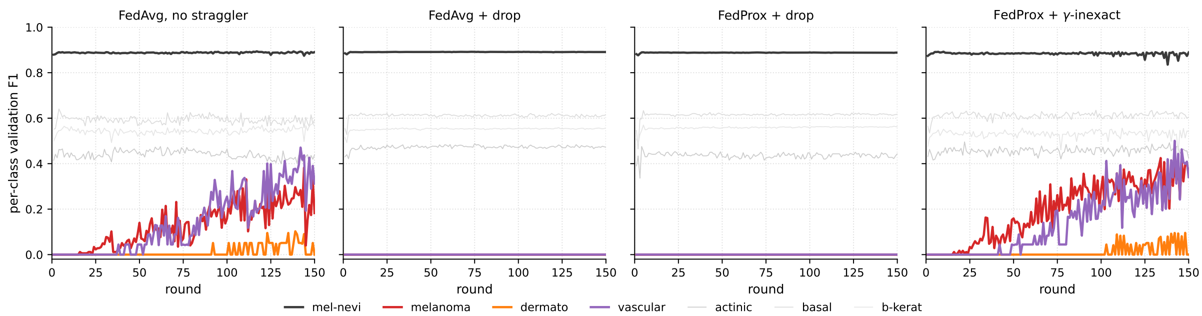

Figure 3.4: Per-class validation F1 trajectories under the four conditions of the L4 decomposition (Table 3.3); mean over three seeds. The majority class (mel-nevi) stays near 0.89 in every cell, so the aggregate improvement is not driven by the dominant class. The three rare classes (dermatofibroma, melanoma, vascular) flatline at zero under *both* drop-handling cells — FedAvg+drop and FedProx+drop alike, so the proximal term alone does not rescue them — and recover only under FedProx+γ-inexact, identifying γ-inexact acceptance, not the proximal term, as the locus of the mechanism.

The perfect-storm experiment stresses the same ordering rather than re-deriving it: on the same L4 setting with a 90% per-round random-straggler rate (batch 10, no momentum), FedAvg+drop flatlines at its round-1 value — best-val selects the untrained init, mean test macro-F1 0.087 ± 0.049 — while FedProx+γ-inexact recovers to 0.365 ± 0.023 at $\mu = 1.0$ and

0.491 ± 0.003 at $\mu = 0.01$, the latter the largest FedProx–FedAvg gap in the thesis ($\Delta = +0.404$; Figure 3.3B). The mechanism is the one the L4 decomposition isolated (Figure 2.6), pushed to its limit; the smaller μ wins because looser anchoring leaves more rare-class learning signal per round. This is a magnitude demonstration, not a second causal decomposition: the schedule and arms differ from the four-condition design, and FedAvg+drop collapses too completely for a clean γ-inexact-vs-proximal split.

Finally, the L1 control tests whether partial-update acceptance is simply a better generic straggler policy. It is not. L1 uses the same four-condition design but with identical label proportions, so the split changes quantity, not class information. All four cells are then nearly flat (Table 3.4): accepting the partial update adds only +0.012, against +0.111 at L4.

Table 3.4: L1 cross-partition control: same four-condition design as Table 3.3 but on the L1 partition (quantity-only heterogeneity, JS divergence ≈ 0); $n = 3$ matched seeds, test macro-F1 mean $\pm$ sample SD with rare-class F1.

Condition	Test macro-F1	Rare-class F1
FedAvg, no straggler	0.601 ± 0.010	0.530 ± 0.008
FedAvg + drop	0.599 ± 0.006	0.539 ± 0.012
FedProx + drop (control)	0.593 ± 0.022	0.537 ± 0.039
FedProx + γ-inexact	0.605 ± 0.009	0.556 ± 0.014

Decomposition (vs L4 in parentheses): proximal-only $= 0.593 - 0.599 = -0.006$ (L4: +0.004); γ-inexact $= 0.605 - 0.593 = +0.012$ (L4: +0.111); drop-cost $= 0.601 - 0.599 = +0.002$ (L4: +0.121). At L1 every rare-F1 cell sits in 0.53–0.56, whereas at L4 the two drop cells fall to zero and only γ-inexact recovers rare-F1 (to 0.29).

The L4/L1 contrast is therefore the key mechanism evidence. At L4, the dropped client carries class information the other client does not have, so accepting its partial update rescues rare-class learning. At L1, the dropped client carries no unique class signal, so there is little to rescue. The straggler mechanism is not generic drift control; it matters when system heterogeneity suppresses statistically unique rare-class information.

The straggler experiments isolate one axis of system asymmetry. Section 3.4 asks whether persistent learning-rate imbalance — a different axis — changes the optimiser ranking.

3.4 Learning-rate asymmetry and FedNova regime-dependence

This experiment tests a different form of system heterogeneity from § 3.3. There, the rare-class client was slow and its partial update could be dropped. Here, the rare-class client is not dropped: it still participates in every round, but its learning rate is made smaller. The question is whether an optimiser can preserve rare-class learning when the rare client's update becomes progressively weaker.

The experiment is run on L4. Client 0 is the dominant client and keeps lr $= 0.01$ throughout. Client 1 holds the rare classes and is assigned one of six smaller learning rates, producing asymmetry ratios from 1:1 to 50:1. Each ratio is tested with FedAvg, FedProx, and FedNova using $n = 3$ matched seeds per cell, giving 54 runs in total. Test macro-F1 across the six-ratio envelope is reported in Table 3.5.

Table 3.5: Learning-rate-asymmetry envelope at L4, $n = 3$ matched seeds per cell. Client 0 (dominant) holds lr = 0.01 throughout; Client 1 (rare-class holder) takes the smaller rate that sets each ratio. Columns are test macro-F1 for the three optimisers and the FedNova advantage over FedAvg and FedProx. FedNova (bold) leads at every ratio and stays flat across the envelope, while FedAvg and FedProx plateau at the rare-class floor by 5:1.

Ratio	Client 0 lr	Client 1 lr	FedAvg	FedProx	FedNova	FN−FA	FN−FP
1:1	0.0100	0.0100	0.457	0.491	**0.509**	+0.052	+0.018
2:1	0.0100	0.0050	0.425	0.462	**0.526**	+0.101	+0.064
5:1	0.0100	0.0020	0.375	0.392	**0.504**	+0.129	+0.112
10:1	0.0100	0.0010	0.369	0.369	**0.501**	+0.132	+0.132
20:1	0.0100	0.0005	0.367	0.372	**0.486**	+0.119	+0.114
50:1	0.0100	0.0002	0.367	0.367	**0.498**	+**0.131**	+**0.131**

The pattern is clear (Table 3.5). FedAvg and FedProx both degrade as soon as Client 1's learning rate is reduced and plateau near 0.367 macro-F1 by 5:1 — the rare-class floor, where the model still learns the dominant-client distribution but rare-class learning has largely collapsed. FedNova behaves differently, holding 0.486–0.526 across the full 1:1–50:1 envelope; even at 50:1 it reaches 0.498 against 0.367 for both FedAvg and FedProx, a +0.131 macro-F1 gap. Rare-class F1 confirms this is not a macro-F1 artefact: it is near zero for FedAvg and FedProx by 10:1 (FA 0.016, FP 0.010) but stays around 0.35 for FedNova across the envelope (Figure 3.5).

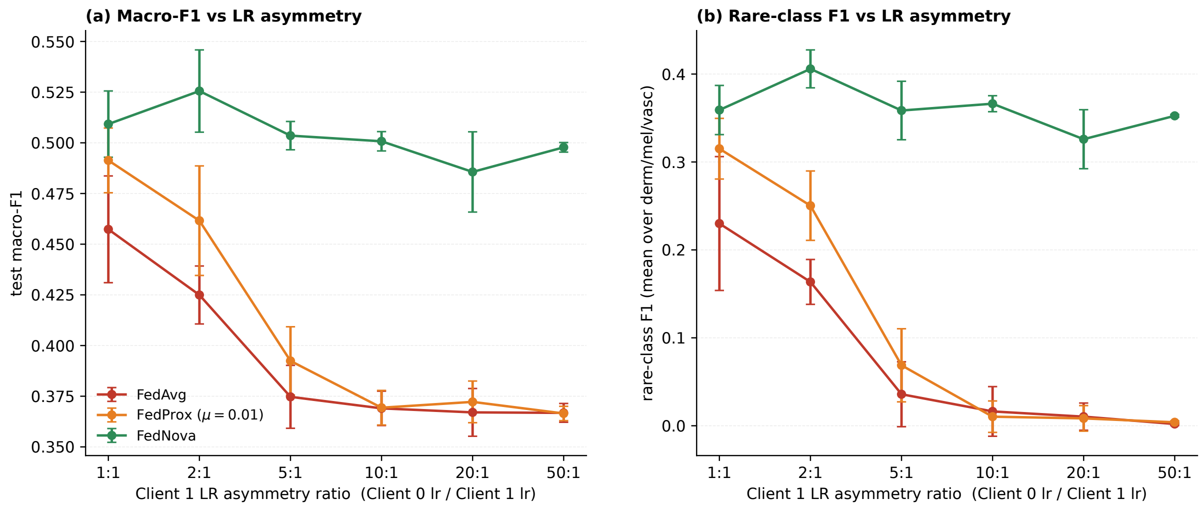

Figure 3.5: Learning-rate asymmetry envelope at L4, $n = 3$ per cell. Two panels: test macro-F1 and rare-class F1 (mean over dermato, melanoma, vascular) vs the Client 1 LR ratio on a $\log_{10}$ axis from 1:1 to 50:1. FedAvg and FedProx plateau at the rare-class floor; FedNova stays flat across the envelope. Exact macro-F1 values and FedNova deltas are in Table 3.5.

The 5:1 cell is where the metric choice earns its keep. Raw accuracy would prefer FedAvg (0.753) over FedNova (0.726), yet FedAvg's macro-F1 (0.375, rare-F1 ≈ 0.02) shows it has lost the rare classes while FedNova has not (0.504, rare-F1 ≈ 0.37): the accuracy gap is small only because both still classify mel-nevi (66.9 % of the test set) correctly. This is why § 2.2 ranks by macro-F1, not accuracy.

The interpretation is that a reduced learning rate makes Client 1's local model move less, so its update carries weaker rare-class information. FedAvg does not correct this, and the FedProx proximal term cannot restore an update that is already too small; FedNova, by normalising updates by effective local work, preserves Client 1's contribution in this deterministic-asymmetry setting.

The update-norm logs are mechanism-consistent: at 5:1 Client 1's update retains 0.91 of its 1:1 size under FedNova but only about half (0.47 FedAvg, 0.50 FedProx), so the rare-client contribution is preserved under normalisation and attenuated without it (§ 3.6).

FedNova's success here is not universal: it fails under random-τ straggler heterogeneity, where the amount of local work varies randomly from round to round. On the same 7-client engineered partition as the § 3.3 screening comparison (Regime A, per-round random schedule; *not* an L4 experiment), FedNova obtains only 0.237 ± 0.094 macro-F1, far below the matching FedAvg and FedProx cells ($n = 10$: FA 0.493 ± 0.028, FP 0.510 ± 0.017). Best-val selection masks the severity: three of ten seeds already sit at the majority-class-only baseline (≈ 0.114, predicting mel-nevi for every image) at best-val. The final validation histories show all ten runs ending below 0.20 macro-F1, with nine of ten reaching the strict mel-nevi-only pattern; because final *test* metrics were not saved for this sweep, this is reported as a validation-trajectory failure mode rather than a final-test estimate.

This contrast is the section's main point: FedNova is empirically robust under deterministic learning-rate asymmetry but brittle under random, noisy local work. The mechanism is work-normalisation: FedNova normalises by effective local work (τ), and this normalised aggregation interacts favourably with a stable, persistently weakened update but amplifies unstable partial updates under random τ; the precise cause is left to future work, so FedNova is best read as regime-dependent rather than universally superior (Table 3.6).

Table 3.6: FedNova across the two system-heterogeneity regimes: persistent deterministic learning-rate asymmetry ($n = 3$) versus random per-round τ stragglers ($n = 10$, outside the stable-τ assumption of Wang et al. (2020)). Test macro-F1; the mechanism reading is given in the text.

FedNova regime	Result	Interpretation
Persistent LR asymmetry	macro-F1 holds 0.49–0.53 across 1:1–50:1	Empirically robust; robustness from work-normalised (τ) aggregation
Random-τ stragglers	macro-F1 $= 0.237 \pm 0.094$; 3/10 majority-only at best-val, 9/10 by the final validation round (no final test saved)	τ-normalisation amplifies noisy partial updates (outside Wang 2020)

Across these regimes, optimiser gaps become large only under specific stress, and the failing protocols fail in a clinically specific way — they lose rare-class sensitivity. The next section shows this failure usually appears as rare-class images routed into the dominant mel-nevi class.

3.5 Rare-class collapse and the loss-side comparison

The previous sections show *when* the optimiser gap becomes large; this section asks what those failures look like at the class level, and whether the FedProx advantage is merely class-imbalance correction in disguise.

Per-class collapse. The low-macro-F1 cells in §§ 3.3–3.4 fail the same way — on the rare classes, not the majority. Synthesising per-class optimiser differences across the whole study, the large FedProx−FedAvg (and FedNova−FedAvg) per-class F1 gaps sit almost entirely in the rare-class columns and almost entirely in the system-heterogeneity cells (Figure 3.6). The failure itself is misrouting: rare-class images are predicted as mel-nevi. Mean rare-to-mel-nevi misrouting (over dermato, melanoma, vascular) ranges from 24 % in the best protocol (perfect-storm FP $\mu = 0.01$, $n = 3$) to 67 % in the worst (perfect-storm FA+drop, $n = 3$), and 28 % even under the IID Flower negative control ($n = 10$) — a class-imbalance saturation that the FL protocol preserves or amplifies but does not itself create. A confusion-matrix view of this rare-to-mel-nevi routing across the four-condition cells is given in Appendix A.4 (Figure A.1).

The D1 5:1 cell isolates the optimiser dimension: melanoma *recall* is 1.20 % (FedAvg), 6.13 % (FedProx), and 46.34 % (FedNova) — the same disease across three regimes spanning two orders of magnitude. Because macro-F1 weights all seven classes equally, losing one class moves it by ≈ 0.057, the order of the entire engineered Δ (§ 3.1). The same shape recurs across aggregation dropout, learning-rate asymmetry, and the $\mu = 1.0$ over-regularisation cell (§ 3.2).

Table 3.7: Rare-class collapse across four representative cells spanning the straggler and learning-rate-asymmetry axes ($n = 3$ each). Columns: Rare-F1 (mean F1 over dermato, melanoma, vascular), melanoma recall, melanoma F1, and rare-to-mel-nevi misrouting. Mel. recall reports whether melanoma is recovered at all; mel. F1 adds the precision penalty.

Protocol	Algorithm	Rare-F1	Mel. recall	Mel. F1	Misrouting	Interpretation
Perfect-storm + drop	FedAvg	0.000	0.00 %	0.00 %	67 %	Rare client dropped; flatline
Perfect-storm + γ	FedProx ($\mu = 0.01$)	0.327	38.27 %	34.46 %	24 %	γ-inex preserves rare signal
D1 5:1 LR asymmetry	FedAvg	0.036	1.20 %	2.28 %	51 %	Rare client under-weighted; gradual erosion
D1 5:1 LR asymmetry	FedNova	0.359	46.34 %	36.74 %	26 %	Robust to the stable weakened update

These four cells (Table 3.7) span instant-collapse and gradual-erosion modes across straggler and LR-asymmetry axes; the misrouting rate co-varies with rare-F1 across all four.

Collapse modes. The collapse takes two temporal forms of one mechanism: an *instant* collapse when updates are dropped wholesale (perfect-storm FedAvg+drop flatlines at its round-1 value, mean 0.087 ± 0.049), and a *gradual* erosion when they are merely under-weighted (D1 5:1 drifts $0.457 \to 0.375$ with rare-F1 falling in lockstep).

Loss-side imbalance correction (weighted CE and focal loss). To test whether the FedProx advantage might be class-imbalance correction in disguise, we ran FedAvg and FedProx at L4 with three loss functions ($n = 3$ per cell): plain cross-entropy, class-weighted CE with inverse-frequency weights, and focal loss ($\gamma = 2$). The results are in Table 3.8.

Table 3.8: Loss-side imbalance correction at L4 ($n = 3$ per cell). Cells are test macro-F1 ± sample SD with rare-class F1 in parentheses; the strongest cell is bold.

Algorithm	CE	Weighted CE	Focal
FedAvg	0.477 ± 0.002 (0.285)	$\mathbf{0.505 \pm 0.023}$ **(0.367)**	0.467 ± 0.002 (0.317)
FedProx	0.478 ± 0.016 (0.291)	0.503 ± 0.014 (0.346)	0.476 ± 0.022 (0.304)

Three points follow (Table 3.8). Weighted CE is the strongest loss-side correction: FedAvg+wCE (0.505) marginally exceeds FedProx+CE (0.478), the CE-to-wCE step alone raising FedAvg rare-F1 by +0.082. Focal raises rare-F1 but lowers macro at $\gamma = 2$. And FedProx adds little on top of either loss — FP+wCE sits inside FA+wCE's seed SD — so FedProx acts as a drift/protocol intervention complementary to direct loss-side correction. This does not conflict with § 3.3: γ-inexact acceptance restores aggregation under straggler-induced signal loss while weighted CE re-weights rare classes during local training, so they act at different pipeline stages and are combinable in principle, though tested only separately here ($n = 3$, L4).

The rare-class collapse documented here is clinically *motivated*, not clinically *validated*: DermaMNIST is a 28×28-pixel benchmark, and external validation on real multi-site workflows remains future work. The next section turns to the per-round update-norm activity behind these per-class outcomes.

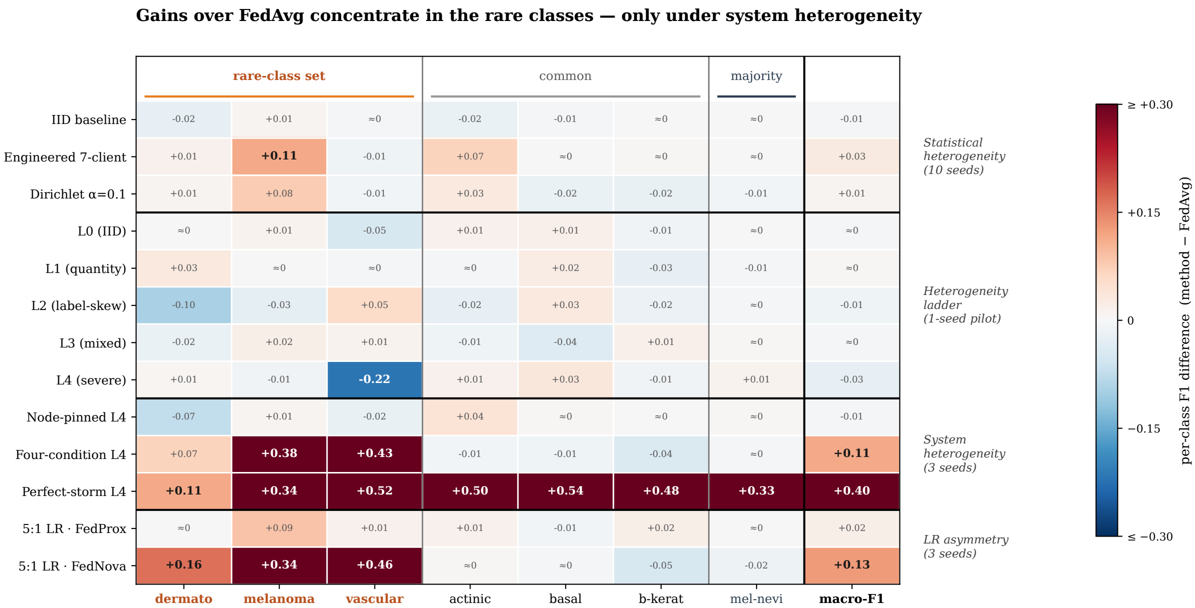

Figure 3.6: Cross-experiment synthesis: per-class FedProx−FedAvg (or FedNova−FedAvg) test-F1 differences across the principal experiment cells — statistical-heterogeneity baselines, the heterogeneity ladder, the four-condition L4 decomposition, perfect-storm, asymmetric LR, and node-pinned L4. The mel-nevi column is near zero everywhere (both optimisers already classify the majority correctly) and the statistical-heterogeneity rows are near zero throughout; large positive differences appear in the rare-class columns (dermatofibroma, melanoma, vascular) precisely in the system-heterogeneity cells.

3.6 Mechanism and robustness

After the main regime contrasts, these diagnostics check whether the drift-control reading is consistent with update-level behaviour and whether the optimiser ranking is robust to reporting choices.

Update-norm activity. Update norms ($\|\Delta w\|$, the size of the client update returned to the server) are used here only as a supporting diagnostic, not as causal proof. Two facts emerge. First, FedProx does what its proximal term intends: on the engineered partition the mean per-round update norm falls from 1.523 (FedAvg) to 1.029 (FedProx, $\mu = 0.01$), a 32% reduction (Figure 3.7; update-norm diagnostic logs were available for 9 FedAvg, 8 FedProx and 7 FedNova seeds, not the full ten). This is a local-training effect and does not contradict § 3.3, where the straggler-regime gain came from accepting the rare client's partial update rather than from the proximal term.

Second, as a separate cross-protocol correlation (not shown in Figure 3.7), update size tracks rare-class performance only within a controlled protocol. Pooled across 78 L4 runs, larger dominant-client update norms correlate with higher rare-class F1 (Spearman $r = +0.634$), but the Client 1-to-Client 0 ratio does not ($r = -0.045$) because the pool mixes mechanisms. Inside the asymmetric-LR experiment alone, rare-class F1 closely follows the rare client's relative update strength ($r = +0.80$ FedAvg, $r = +0.87$ FedProx). These are mechanism-consistent correlations; the main claims rest on the controlled contrasts in § 3.3 and § 3.4.

Protocol-flip audit. The protocol-flip audit checks whether the algorithm ranking depends on the reporting checkpoint (best-validation vs final round vs last-K mean); a flip is a sign change of the paired difference across these choices. Large effects do not flip and small ones do: 0/9 flips in the large-effect settings (Li-style decomposition, perfect-storm) versus 3/9 in the small-effect settings. The centrepiece § 3.3 effect (+0.115) and the § 3.4 FedNova gap (+0.131) therefore survive checkpoint choice, while the small § 3.1 separations should be read as fragile. Several further diagnostics (the B-dissimilarity proxy, training-instability metric, asymmetric-μ probe, early-warning detector, and small-hospital case study) were examined but are not load-bearing. Both checks support the same message: large regime-dependent effects are robust, small symmetric-protocol differences are not.

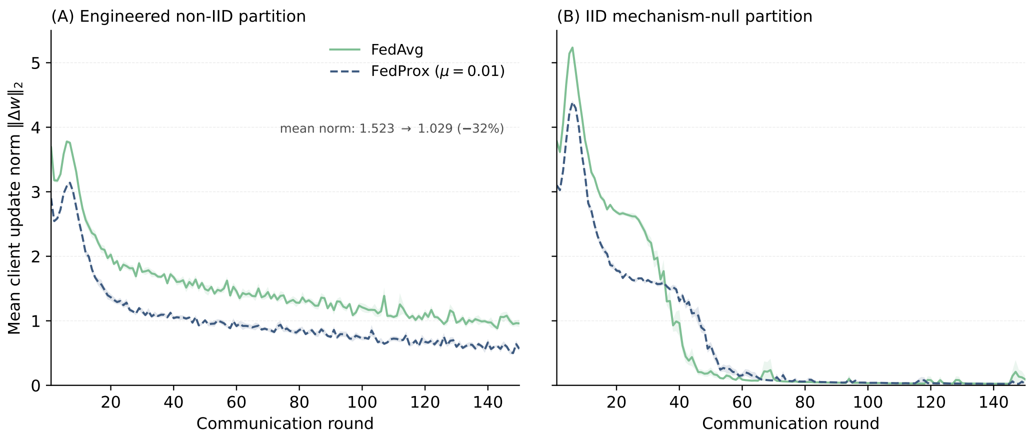

Figure 3.7: Update-norm mechanism diagnostic. Mean per-round client update-norm ($\|\Delta w\|_2$) trajectories for FedAvg and FedProx on the engineered non-IID partition (panel A) and the IID mechanism-null partition (panel B), with light ± 1 SEM bands across seeds. Update-norm logs were available for 9 FedAvg and 8 FedProx seeds on the engineered partition (10 each on the IID partition). FedProx tracks lower update norms under the engineered partition, consistent with the proximal term's drift-control role; on the IID partition the two trajectories sit low and close together, as expected when the mechanism is largely inactive. This is a supporting diagnostic.

4 Discussion and Conclusion

4.1 Discussion

The results are best read not as a contest between three optimisers but as a map of when each preserves rare-class signal in the global model. Optimiser performance is regime-dependent for one reason: rare-class performance depends on whether the rare client's contribution survives aggregation, and different optimisers protect or forfeit it under different heterogeneity. This single rule organises the six experiments across the regimes studied.

Read through that rule, the findings cohere. Statistical heterogeneity alone separates FedAvg and FedProx only slightly and runtime-sensitively (§3.1), too fragile to justify a universal preference, and the proximal coefficient is sharply tuned: useful near $\mu = 0.01$ but harmful at $\mu = 1.0$, where it damages exactly the rare classes (§3.2). This already indicates that the proximal term acts on local drift, not on whether a client's signal is admitted.

The L4/L1 decomposition makes the point precise (§3.3): under straggler asymmetry the FedProx advantage of about +0.115 macro-F1 comes almost entirely (+0.111) from γ-inexact acceptance of the straggler's partial update and only +0.004 from the proximal term, while the matched L1 control — where the suppressed client carries nothing unique — collapses the effect, confirming that admitting the partial update matters only when that client is the sole source of rare-class signal.

FedNova fits the same picture from another angle (§3.4): it is empirically robust under deterministic learning-rate asymmetry (an advantage of about +0.131 macro-F1 at the most extreme ratio) but amplifies noise and collapses to roughly 0.237 under random, round-to-round straggling — a robustness best read as its work-normalised (τ) aggregation interacting favourably with a *stable* weakened update, not as direct learning-rate normalisation.

Finally, class-weighted cross-entropy matches or exceeds FedProx with plain cross-entropy on the symmetric partition (§3.5), locating imbalance correction as a loss-side layer distinct from the aggregation-and-protocol layer where FedProx acts; the two address different failure modes and could in principle be combined. This loss-stage comparison does not explain away the separate γ-inexact straggler result, which acts at the protocol/aggregation stage.

These findings refine rather than contradict prior work. FedAvg (McMahan et al., 2017) behaves as expected when clients are well matched; for FedProx (Li et al., 2020), the decomposition indicates — within this benchmark and this simulated protocol — that the straggler-regime benefit is carried mainly by inexact-update acceptance rather than by the proximal term, an attribution to the FedProx-plus-γ-inexact protocol as configured here rather than a claim that the method underperforms; and for FedNova (Wang et al., 2020) the random-τ collapse is consistent with operating outside its stable-work assumption, not with a flaw in its analysis. No result here should be read as refuting these analyses. The clinical reading stays cautious: rare lesions collapsing into melanocytic nevi is the shared failure mode, but DermaMNIST is a 28×28 benchmark, so this is clinically motivated, not validated.

The headline findings and supporting diagnostics, with explicit strength labels, are collected in Table 4.1.

Table 4.1: Final claim–evidence summary. Strength labels: *strong* = multi-seed, mechanism-isolable, checkpoint-robust; *medium* = multi-seed but small effect or single-protocol; *exploratory* = correlational, single-protocol, or $n = 3$ without mechanism isolation.

Claim	Evidence	Strength	Implication
FedProx not free under IID	Flower IID $\Delta = -0.007$ ($n = 10$)	strong	Reach for FedProx only when heterogeneity exists
Stat-het alone gives fragile separation	Flower engineered $\Delta = +0.007$ (7/10 wins, $n = 10$)	medium	Do not recommend FedProx on partition alone
μ must be tuned	Sweep $n = 10$: peak $+0.013$ at $\mu = 0.01$; -0.074 at $\mu = 1.0$	strong	Default $\mu = 0.01$; do not raise under heterogeneity
γ-inexact dominates straggler gain	L4 four-condition $n = 3$: total $+0.115$, γ-inex $+0.111$; L1 control $+0.012$	medium–strong	Use FedProx + γ-inexact under straggler protocols
FedNova regime-dependent	LR envelope: FN holds 0.49–0.53 across 1:1–50:1 ($n = 3$); random-τ collapses to 0.237 ($n = 10$; 9/10 majority-only by the final validation round)	medium–strong	Prefer FedNova under persistent LR asymmetry, not under random-τ
Rare-class collapse is the clinically relevant benchmark failure mode	Misrouting 24 %–67 %; D1 5:1 melanoma recall = 1.2/6.1/46.3 % (F1 2.3/10.8/36.7 %, $n = 3$)	strong	Report per-class metrics; macro-F1 alone hides it
wCE competes with FedProx	L4 $n = 3$: FA+wCE 0.505 vs FP+CE 0.478	medium	Loss-side rebalancing is a competitive intervention
Update-norms support drift-control reading	Engineered: FA 1.523 vs FP 1.029 (-32 %); cross-protocol $r = +0.634$	exploratory	Update norms corroborate the drift-control reading

4.2 Limitations

Several scope conditions bound these conclusions. First, the study is a controlled benchmark simulation rather than a clinical deployment. All experiments use DermaMNIST at 28×28 resolution, stragglers are simulated through local-epoch budgets rather than wall-clock deadlines, and the L1/L4 partitions are stylised stress regimes designed to isolate mechanism behaviour. These choices make the failure modes easier to attribute, but they also mean that the results should be read as a regime map for controlled heterogeneity settings, not as a validated claim about real hospital networks or full-resolution dermatology workflows.

Second, the evidence is strongest where the experiments are replicated and more exploratory where compute limited the number of seeds. The headline comparisons use larger seed counts, while the mechanism probes use three matched seeds and are reported as directional rather than significance-tested. The heterogeneity ladder is a single-seed exploratory pilot. In addition, the engineered partition shows the same direction of FedProx–FedAvg separation across the pure-PyTorch and Flower runtimes, but with different magnitudes, so the robust conclusion is directional rather than an exact effect-size claim.

Third, the optimiser and evaluation space is deliberately limited. The comparison covers FedAvg, FedProx, and FedNova, but not variance-reduced or adaptive federated methods such as SCAFFOLD (Karimireddy et al., 2020), FedAdam, or FedYogi (Reddi et al., 2021). Evaluation is based on argmax predictions, macro-F1, and per-class diagnostics; AUC, calibration, and cross-client fairness measures are not reported.

These limitations mainly restrict the breadth of the claims rather than the internal comparison. The thesis does not establish a universal optimiser ranking or a clinically validated recommendation. It instead identifies where rare-class signal is preserved or lost under controlled forms of statistical and system heterogeneity. Directions for addressing these constraints are set out in the Conclusion.

4.3 Conclusion

This thesis asked not whether a federated optimiser is best for imbalanced medical image classification but *when* each preserves rare-class performance. The answer is a regime map governed by a single rule: rare-class performance is preserved when rare-client signal reaches the global model. Statistical heterogeneity alone yields only a fragile FedProx–FedAvg separation (RQ1); under straggler asymmetry the FedProx advantage is carried mainly by γ-inexact partial-update handling rather than the proximal term, and only when the suppressed client holds unique rare-class signal (RQ2); FedNova is regime-dependent, robust under deterministic learning-rate asymmetry but brittle under random stragglers (RQ3); and rare-class collapse into melanocytic nevi is the common failure mode, with weighted cross-entropy a competitive loss-side alternative for rare-class preservation rather than evidence that the γ-inexact straggler effect is class-imbalance correction in disguise (RQ4). Concretely, in the perfect-storm stress this margin is large: FedProx with γ-inexact acceptance reaches 0.491 test macro-F1 where FedAvg with straggler dropping collapses to 0.087 — the thesis's largest FedAvg–FedProx gap, $\Delta = +0.404$. Future work should combine γ-inexact aggregation with a weighted-cross-entropy local objective, which were tested only separately here; validate the map on higher-resolution and additional datasets with real wall-clock straggler protocols; add variance-reduced and adaptive optimisers; and log per-client validation to quantify fairness across sites. In imbalanced medical federated learning, optimiser choice should be driven by the heterogeneity regime: the best method is the one whose protocol keeps rare-client signal reaching the global model.

Bibliography

Daniel J. Beutel, Taner Topal, Akhil Mathur, Xinchi Qiu, Titouan Parcollet, Pedro Porto Buarque de Gusmão, and Nicholas D. Lane. Flower: A friendly federated learning research framework, 2020.

Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

Kevin Hsieh, Amar Phanishayee, Onur Mutlu, and Phillip B. Gibbons. The non-IID data quagmire of decentralized machine learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pages 4387–4398. PMLR, 2020.

Tzu-Ming Harry Hsu, Hang Qi, and Matthew Brown. Measuring the effects of non-identical data distribution for federated visual classification, 2019.

Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. Advances and open problems in federated learning. *Foundations and Trends in Machine Learning*, 14(1–2):1–210, 2021. doi: 10.1561/2200000083.

Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank J. Reddi, Sebastian U. Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pages 5132–5143. PMLR, 2020.

Qinbin Li, Yiqun Diao, Quan Chen, and Bingsheng He. Federated learning on non-IID data silos: An experimental study. In *Proceedings of the 38th IEEE International Conference on Data Engineering (ICDE)*, pages 965–978. IEEE, 2022.

Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 2, pages 429–450, 2020.

Xiaoxiao Li, Meirui Jiang, Xiaofei Zhang, Michael Kamp, and Qi Dou. FedBN: Federated learning on non-IID features via local batch normalization. In *International Conference on Learning Representations (ICLR)*, 2021.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings* *of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 54 of *Proceedings of Machine Learning Research*, pages 1273–1282. PMLR, 2017.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, pages 8024–8035. Curran Associates, Inc., 2019.

Sarthak Pati, Ujjwal Baid, Brandon Edwards, Micah Sheller, Shih-Han Wang, G. Anthony Reina, Patrick Foley, Alexey Gruzdev, Deepthi Karkada, Christos Davatzikos, Chiharu Sako, Satyam Ghodasara, Michel Bilello, Suyash Mohan, Philipp Vollmuth, Gianluca Brugnara, Chandrakanth J. Preetha, Felix Sahm, Klaus Maier-Hein, Maximilian Zenk, Martin Bendszus, Wolfgang Wick, Evan Calabrese, Jeffrey Rudie, Javier Villanueva-Meyer, Soonmee Cha, Madhura Ingalhalikar, Manali Jadhav, Umang Pandey, Jitender Saini, John Garrett, Matthew Larson, Robert Jeraj, Stuart Currie, Russell Frood, Kavi Fatania, Raymond Y. Huang, Ken Chang, Carmen Balana, Jaume Capellades, Josep Puig, Johannes Trenkler, Josef Pichler, Georg Necker, Andreas Haunschmidt, Stephan Meckel, Gaurav Shukla, Spencer Liem, Gregory S. Alexander, Joseph Lombardo, Joshua D. Palmer, Adam E. Flanders, Adam P. Dicker, Haris I. Sair, Craig K. Jones, Archana Venkataraman, Meirui Jiang, Tyler Y. So, Cheng Chen, Pheng Ann Heng, Qi Dou, Michal Kozubek, Filip Lux, Jan Michálek, Petr Matula, Miloš Jeřábek, Victor Sanchez, Hassan Vallejo, Ikram Brahim, Sorina Camarasu-Pop, Tristan Glatard, Michel Bilello, Ekaterina Kondrateva, Wilbur Brendel, Malte D. Lücken, Tine Mušic, Carlos A. Ferreira, M. Jorge Cardoso, Marc Modat, Hugo Kuijf, Esther E. Bron, Seyed Sadegh Mohseni Salehi, Alexandros Karargyris, Tabea M. Anthony, Marius George Linguraru, Sai Praneeth Karimireddy, Lalitha Krishnan, John Mongan, Christopher Hess, Soonmee Cha, Javier Villanueva-Meyer, John B. Freymann, Justin S. Kirby, Benedikt Wiestler, Priscila Crivellaro, Christos Davatzikos, Keyvan Farahani, Jayashree Kalpathy-Cramer, Aikaterini Kotrotsou, Dominic Labella, Bjoern Menze, John T. Mongan, Russell T. Shinohara, Roland Wiest, and Spyridon Bakas. Federated learning enables big data for rare cancer boundary detection. *Nature Communications*, 13(1):7346, 2022. doi: 10.1038/s41467-022-33407-5.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Sashank J. Reddi, Zachary Charles, Manzil Zaheer, Zachary Garrett, Keith Rush, Jakub Konečný, Sanjiv Kumar, and H. Brendan McMahan. Adaptive federated optimization. In *International Conference on Learning Representations (ICLR)*, 2021.

Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletarì, Holger R. Roth, Shadi Albarqouni, Spyridon Bakas, Mathieu N. Galtier, Bennett A. Landman, Klaus Maier-Hein, Sébastien

Ourselin, Micah Sheller, Ronald M. Summers, Andrew Trask, Daguang Xu, Maximilian Baust, and M. Jorge Cardoso. The future of digital health with federated learning. *npj Digital Medicine*, 3(1):119, 2020. doi: 10.1038/s41746-020-00323-1.

Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, Mikhail Milchenko, Weilin Xu, Daniel Marcus, Rivka R. Colen, and Spyridon Bakas. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10(1):12598, 2020. doi: 10.1038/s41598-020-69250-1.

Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5: 180161, 2018. doi: 10.1038/sdata.2018.161.

Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.

Jianyu Wang, Qinghua Liu, Hao Liang, Gauri Joshi, and H. Vincent Poor. Tackling the objective inconsistency problem in heterogeneous federated optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 7611–7623, 2020.

Lixu Wang, Shichao Xu, Xiao Wang, and Qi Zhu. Addressing class imbalance in federated learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(11):10165–10173, 2021. doi: 10.1609/aaai.v35i11.17219.

Yuxin Wu and Kaiming He. Group normalization. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 3–19, 2018.

Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2—a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023. doi: 10.1038/s41597-022-01721-8.

Muhammad Mateen Yaqoob, Musleh Alsulami, Muhammad Amir Khan, Deafallah Alsadie, Abdul Khader Jilani Saudagar, and Mohammed AlKhathami. Federated machine learning for skin lesion diagnosis: An asynchronous and weighted approach. *Diagnostics*, 13(11):1964, 2023. doi: 10.3390/diagnostics13111964.

Yue Zhao, Meng Li, Liangzhen Lai, Naveen Suda, Damon Civin, and Vikas Chandra. Federated learning with non-IID data, 2018.

A Provenance, reproducibility, and supplementary diagnostics

A.1 Code, data, and environment

Code availability. All code accompanies this submission as the `fl_dermamnist` repository, released under the MIT licence (`LICENSE`); its top-level `README.md` maps each Results section to the code, raw results, and figure that support it, together with the commands below.

Data availability. Experiments use DermaMNIST from MedMNIST v2 (Yang et al., 2023), derived from HAM10000 (Tschandl et al., 2018). The data is not versioned with the code: it is stored locally as a 64×64 array (`dermamnist_64.npz`) and the loader resizes to the 28×28 resolution used throughout. It is freely re-obtainable through the pinned `medmnist` package; HAM10000 is released under CC BY-NC 4.0.

Environment. Dependencies are pinned in `requirements.txt` (tested on Python 3.14); the package installs with `pip install -e .` from the repository root.

Reproduction. The report's tables and figures rebuild from the saved per-run artefacts without re-training. `bash infra/local/commands.sh sanity` runs the implementation guards (the FedProx $\mu = 0 \equiv$ FedAvg equivalence, the proximal term, the FedNova normaliser, and straggler handling), and `bash infra/local/commands.sh analyse` regenerates the thesis tables and figures. Full hyperparameter sweeps were run on HPC via the `infra/slurm/submit_*.sh` launchers and are costly to re-run; the saved outputs are the intended reproduction path.

Provenance. Every reported number traces to a raw artefact through the result-traceability matrix (claim $\rightarrow$ section $\rightarrow$ raw files $\rightarrow$ generating script $\rightarrow$ HPC job) and a numerical verification ledger, both under `docs/provenance/`.

A.2 Runtimes and implementation provenance

The implementation modules (aggregation, the pure-PyTorch and Flower training loops under `core/` and `runtimes/`, the partitioner, and the FedNova/straggler strategies) and the unit-test guards are documented in the repository README. The guards cover the proximal term, the $\mu = 0 \Leftrightarrow$ FedAvg equivalence, the FedNova normaliser, straggler dropping, partition determinism, and provenance fields.

Flower production runs save per-round validation histories, selected-round metrics with a provenance dictionary (git commit, hostname, framework and Python/PyTorch versions, timestamps), argmax prediction NPZs (used for confusion matrices and misrouting in ğ3.5), and optional update-norm logs. Older pure-PyTorch headline runs retain histories and metrics but lack git-commit provenance, prediction NPZs, and update-norm logs, which is why the chapter treats them as direction-only corroboration. Every numerical claim in Chapter 3 traces to raw artefacts or freshly regenerated summaries, never to stale committed CSVs.

A.3 Supplementary tables

These tables are referenced from the main text but relocated here to keep the body focused on headline evidence; no values differ from the main-text claims.

Table A.1: Reporting protocol and metric hierarchy. Applied uniformly across the Results sections.

Component	Choice	Reason
Primary metric	Test macro-F1 at best-val checkpoint	Class-imbalanced; accuracy saturates at majority floor
Secondary metrics	Per-class F1, per-class recall (confusion-matrix diagonal), rare-F1, rare-to-mel-nevi misrouting	Surface per-class failure that macro can mask; misrouting names the specific collapse mode
Rare-class definition	$\{$dermato, melanoma, vascular$\}$	Low support + high clinical priority (melanoma is mid-support but high-stakes)
$n = 10$ protocol	Paired sign count + Wilcoxon where applicable	Headline statistical-heterogeneity comparisons
$n = 3$ protocol	Directional only; no significance test	Mechanism probes; power insufficient for p-values
$n = 1$ protocol	Labelled “pilot” (single seed)	Heterogeneity ladder; single-class seed events possible
Runtimes	Flower (central), pure-PyTorch (corroboration)	Flower carries full git+update-norm provenance

Table A.2: Four-condition decomposition design (L4, two-client, Regime B, $E_{\text{straggler}} = 5$ on Client 1, $n = 3$ matched seeds per cell). Holding the partition and schedule fixed across all four cells isolates the optimiser-vs-protocol contribution. The design omits the FedAvg + γ-inexact cell, so the γ-inexact attribution is to the FedProx + γ-inexact *protocol* rather than to γ-inexact acceptance as a standalone universal mechanism.

Condition	Straggler schedule	Partial-update handling	Proximal term	Mechanism isolated
FedAvg, no straggler	uniform ($E = 20$ everywhere)	not applicable	–	No-system-heterogeneity baseline
FedAvg + drop	fixed, Client 1 $E = 5$	drop	–	Cost of dropping the rare-class client’s partial update
FedProx + drop	fixed, Client 1 $E = 5$	drop	$\mu = 0.01$	Proximal-only contribution under matched drop handling
FedProx + γ-inexact	fixed, Client 1 $E = 5$	accept (γ-inex)	$\mu = 0.01$	Additional effect of accepting the partial update

Table A.3: Per-class test F1 mean (at the best-validation checkpoint) across the μ sweep on the engineered partition, $n = 10$ paired seeds per cell. SDs across seeds range 0.003 (mel-nevi) to 0.113 (rare classes). "Δ at $\mu = 1.0$" is the absolute per-class F1 deficit at the catastrophic cell relative to the FedAvg baseline ($\mu = 0$). The deficit concentrates in rare and minority classes; the dominant mel-nevi class is essentially invariant. Source: `mu_sensitivity_flower/`.

Class	$\mu = 0$	$\mu = 0.001$	$\mu = 0.01$	$\mu = 0.1$	$\mu = 1.0$	Δ at $\mu = 1.0$
actinic	0.475	0.470	0.489	0.434	0.351	−0.124
basal	0.497	0.513	0.497	0.405	0.438	−0.060
b-kerat	0.405	0.400	0.423	0.416	0.328	−0.077
dermato	0.259	0.287	0.283	0.279	0.158	**−0.101**
melanoma	0.244	0.258	0.304	0.287	0.216	−0.028
mel-nevi	0.884	0.885	0.886	0.886	0.867	−0.017
vascular	0.687	0.713	0.661	0.646	0.574	**−0.113**

A.4 Supplementary diagnostics

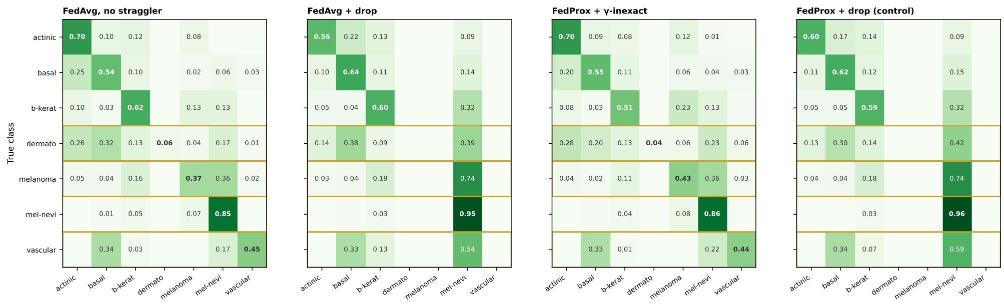

Figure A.1: Confusion-matrix diagnostic for the four-condition L4 decomposition (row-normalised, mean over three seeds; rare classes outlined). Under drop handling (FedAvg + drop and FedProx + drop), rare-class examples — melanoma especially — are frequently routed into the majority mel-nevi class, the misrouting behind the rare-class macro-F1 collapse discussed in §3.5. FedProx + γ-inexact acceptance restores the rare-class diagonals most clearly, consistent with the per-class recovery in §3.3. This figure is diagnostic only; the main quantitative claims use macro-F1 and per-class F1.